2.1

(a)

x	y	$x-\bar{x}$	$(x-\bar{x})^2$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\Sigma x = 5$	$\Sigma y = 10$	$\Sigma(x-\bar{x}) = 0$	$\Sigma(x-\bar{x})^2 = 10$	$\Sigma(y-\bar{y}) = 0$	$\Sigma(x-\bar{x})(y-\bar{y}) = 8$

$\bar{x} = 1 \quad \bar{y} = 2$

(b)

$$b_2 = \frac{\Sigma(x_i-\bar{x})(y_i-\bar{y})}{\Sigma(x_i-\bar{x})^2} = \frac{8}{10} = 0.8$$

$$b_1 = \bar{y} - b_2\bar{x} = 2 - 0.8 \times 1 = 1.2$$

b_2: When x increases 1 unit, y will increase 0.8 unit.

b_1: when $x=0$, $y=1.2$

(c)

$$\sum_{i=1}^{5} x_i^2 = 3^2 + 2^2 + 1^2 + (-1)^2 + 0 = 15$$

$$\sum_{i=1}^{5} (x_i - \bar{x})^2 = 10 = \sum_{i=1}^{5} x_i^2 - N\bar{x}^2 = 15 - 5 \cdot 1^2 = 10$$

$$\sum_{i=1}^{5} x_i y_i = 12 + 4 + 3 - 1 + 0 = 18$$

$$\sum_{i=1}^{5} (x_i - \bar{x})(y_i - \bar{y}) = 8 = \sum_{i=1}^{5} x_i y_i - N\bar{x}\bar{y} = 18 - 5 \cdot 1 \cdot 2 = 8$$

(d)

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i = 5$	$\sum y_i = 10$	$\sum \hat{y}_i = 10$	$\sum \hat{e}_i = 0$	$\sum \hat{e}_i^2 = 3.6$	$\sum x_i \hat{e}_i = 0$

$$\hat{y}_i = b_1 + b_2 x_i = 1.2 + 0.8 x_i$$

(d)

$$S_y^2 = \frac{\sum_{i=1}^{N}(y_i - \bar{y})^2}{N-1} = \frac{10}{4} = 2,5$$

$$S_x^2 = \frac{\sum_{i=1}^{N}(x_i - \bar{x})^2}{N-1} = \frac{10}{4} = 2,5$$

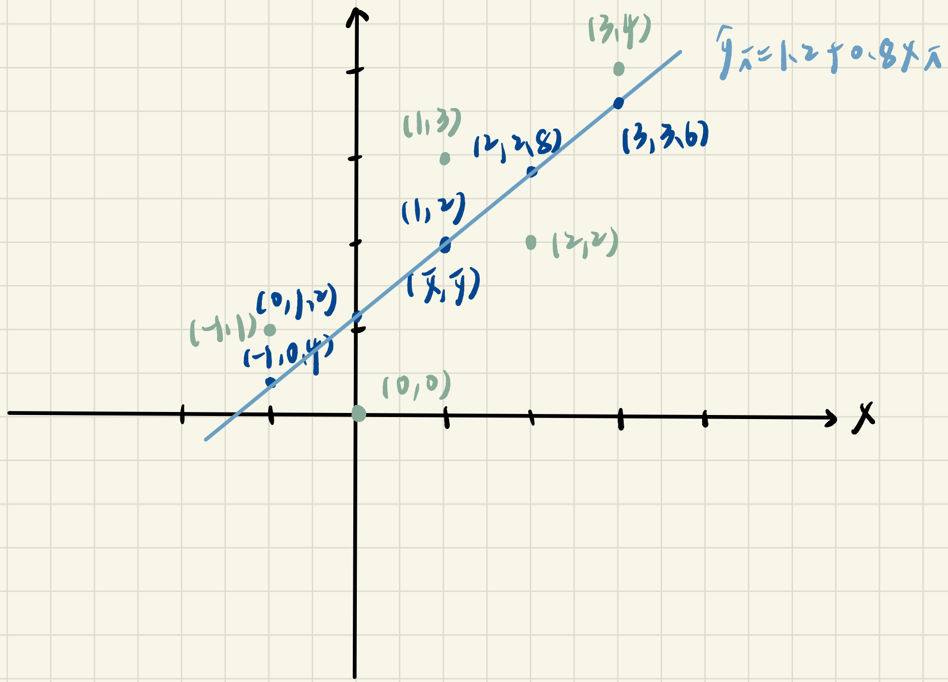
$$S_{xy} = \frac{\sum_{i=1}^{N}(y_i - \bar{y})(x_i - \bar{x})}{N-1} = \frac{8}{4} = 2$$

$$r_{xy} = \frac{S_{xy}}{S_x S_y} = \frac{2}{\sqrt{2,5}\sqrt{2,5}} = 0,8$$

$$CV_x = 100 \cdot \frac{S_x}{\bar{x}} = 100 \cdot \frac{\sqrt{2,5}}{1} = 100\sqrt{2,5} = 158.11388$$

median of $x = 1$

(e)



(f) From (e), fitted line pass through $(\bar{x},\bar{y}) = (1,2)$

(g) $\bar{y} = 1.2 + 0.8 \times 1 = 2$

(h) $\sum \hat{y}_i = 10$ $\quad \bar{\hat{y}}_i = \frac{10}{5} = 2 = \bar{y}$

(i) $\hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} = \frac{3.6}{3} = 1.2$

(j) $Var(b_2|x) = \frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2} = \frac{1.2}{10} = 0.12$

$se(b_2) = \sqrt{0.12} = 0.34641$

(a)

2122

```
Call:
lm(formula = totalscore ~ small, data = data_a)

Residuals:
    Min      1Q  Median      3Q     Max 
-189.44  -50.44   -9.44   43.06  324.38 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  916.442      3.675 249.396   <2e-16 ***
small         12.175      5.369   2.268   0.0236 *  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 74.59 on 773 degrees of freedom
Multiple R-squared:  0.006608,	Adjusted R-squared:  0.005323 
F-statistic: 5.142 on 1 and 773 DF,  p-value: 0.02363
```

顯著

$\widehat{totalscore} = 916.442 + 12.175 \times small$

$\beta_1 = 916.442$，當 small = 0 時，regular-sized class 的預期 totalscore = 916.442 分

$\beta_2 = 12.175$，表示 small class 的 totalscore 比 regular-sized class 高出 12.175 分，且結果統計顯著

由此可知，班級規模對於學生的總成績有統計上顯著的影響，small class 的學生平均而言總成績高於 regular-sized class

(b)

```
Call:
lm(formula = readscore ~ small, data = data_a)

Residuals:
    Min      1Q  Median      3Q     Max 
-74.665 -21.590  -2.665  16.410 194.335 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  432.665      1.488 290.760  < 2e-16 ***
small          6.924      2.174   3.185  0.00151 ** 
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 30.2 on 773 degrees of freedom
Multiple R-squared:  0.01295,	Adjusted R-squared:  0.01167 
F-statistic: 10.14 on 1 and 773 DF,  p-value: 0.001507
```

```
Call:
lm(formula = mathscore ~ small, data = data_a)

Residuals:
     Min       1Q   Median       3Q      Max 
-144.777  -35.028   -5.028   29.223  142.223 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  483.777      2.449 197.545   <2e-16 ***
small          5.251      3.578   1.467    0.143    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 49.71 on 773 degrees of freedom
Multiple R-squared:  0.002778,	Adjusted R-squared:  0.001488 
F-statistic: 2.153 on 1 and 773 DF,  p-value: 0.1427
```

① $\widehat{readscore} = 432.665 + 6.924 \times small$

$\beta_1 = 432.665$：當 small=0 時, regular-sized class 的預期 readscore = 432.665分

$\beta_2 = 6.924$：表示 small class 的 readscore 比 regular-sized class 高出 6.924分, 且統計結果顯著.

② $\widehat{mathscore} = 483.777 + 5.251 \times small$

$\beta_1 = 483.777$：當 small=0 時, regular-sized class 的預期 mathscore = 483.777 分

$\beta_2 = 5.251$：表示 small class 的 readscore 比 regular-sized class 高出 5.251 分, 且統計結果不顯著.

⇒ 班級規模對於成績的影響會因科目而不同. 小班模顯著提高閱讀成績, 但對於數學成績的提升不顯著.

(c)

```
Call:
lm(formula = totalscore ~ aide, data = data_b)

Residuals:
     Min       1Q   Median       3Q      Max 
-191.748  -50.442   -5.442   40.558  312.558 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  916.442      3.559 257.529   <2e-16 ***
aide           4.306      4.994   0.862    0.389    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 72.23 on 835 degrees of freedom
Multiple R-squared:  0.0008898,	Adjusted R-squared:  -0.0003068 
F-statistic: 0.7436 on 1 and 835 DF,  p-value: 0.3888
```

$\widehat{totalscore} = 916.442 + 4.306 \times aide$

$r_1 = 916.442$，當 aide=0，totalscore = 916.442

$r_2 = 4.306$，表示 regular-sized class with teacher aide 的 totalscore 比 regular-sized class 高出 4.306 分，然統計結果不顯著。

由此可知，是否有助教對於 regular-sized class 的 totalscore 提高統計上沒有顯著的影響。

(d)

```
Call:
lm(formula = readscore ~ aide, data = data_b)

Residuals:
    Min      1Q  Median      3Q     Max
-74.665 -21.536  -2.536  15.464 194.335

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  432.665      1.460 296.341   <2e-16 ***
aide           2.871      2.049   1.401    0.161
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 29.64 on 835 degrees of freedom
Multiple R-squared:  0.002347,	Adjusted R-squared:  0.001152
F-statistic: 1.964 on 1 and 835 DF,  p-value: 0.1615
```

```
Call:
lm(formula = mathscore ~ aide, data = data_b)

Residuals:
     Min       1Q   Median       3Q      Max
-146.212  -31.212   -5.777   29.223  142.223

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  483.777      2.355 205.464   <2e-16 ***
aide           1.435      3.304   0.434    0.664
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 47.79 on 835 degrees of freedom
Multiple R-squared:  0.0002258,	Adjusted R-squared:  -0.0009715
F-statistic: 0.1886 on 1 and 835 DF,  p-value: 0.6642
```

$\widehat{readscore} = 432.665 + 2.871 \times aide$

$r_1 = 432.665$. aide = 0. 表示 regular-sized class 的平均 readscore = 432.665 分

$r_2 = 2.871$: 表示 regular-sized class with teacher aide 的 readscore 比 regular-sized class 高出 2.871 分. 統計結果不顯著.

$\widehat{mathscore} = 483.777 + 1.435 \times aide$

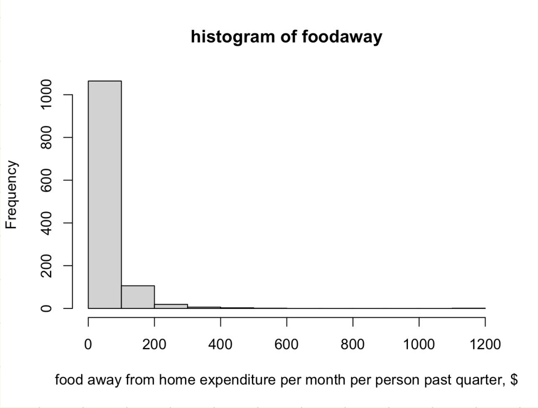
$r_1 = 483.777$ aide = 0. 表示 regular-sized class 的平均 mathscore = 483.777 分

$r_2 = 1.435$. 表示 regular-sized class with teacher aide 的 mathscore 比 regular-sized class 高出 1.435 分. 統計結果不顯著.

⇒ 由此可知, 是否有助教對於 regular-sized class 的 readscore 和 mathscore 的提高統計上都沒有顯著影響.

2.25

(a)

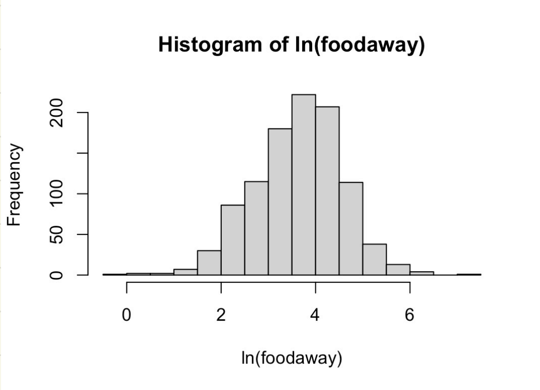



```
> summary(cex5_small$foodaway)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
   0.00   12.04   32.55   49.27   67.50 1179.00 
> quantile(cex5_small$foodaway, probs = c(0.25, 0.75), na.rm = TRUE)
    25%     75% 
12.0400 67.5025 
```

(b)

mean_adv	73.1549416342412
mean_col	48.5971815718157
mean_none	39.0101742160279
med_adv	48.15
med_col	36.11
med_none	26.02

(c)



因為有178個家庭外出用餐的費用是0元，ln(0)是未定義所以取對數之後，0的觀測值會變成missing value，所以ln(foodaway)的有效樣本數會少於foodaway.

```
> summary(cex5_small$ln_foodaway)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's 
-0.3011  3.0759  3.6865  3.6508  4.2797  7.0724     178 
```

(d)

```
Call:
lm(formula = ln_foodaway ~ income, data = cex5_small)

Residuals:
    Min      1Q  Median      3Q     Max 
-3.6547 -0.5777  0.0530  0.5937  2.7000 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 3.1293004  0.0565503   55.34   <2e-16 ***
income      0.0069017  0.0006546   10.54   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.8761 on 1020 degrees of freedom
  (178 observations deleted due to missingness)
Multiple R-squared:  0.09826,	Adjusted R-squared:  0.09738 
F-statistic: 111.1 on 1 and 1020 DF,  p-value: < 2.2e-16
```

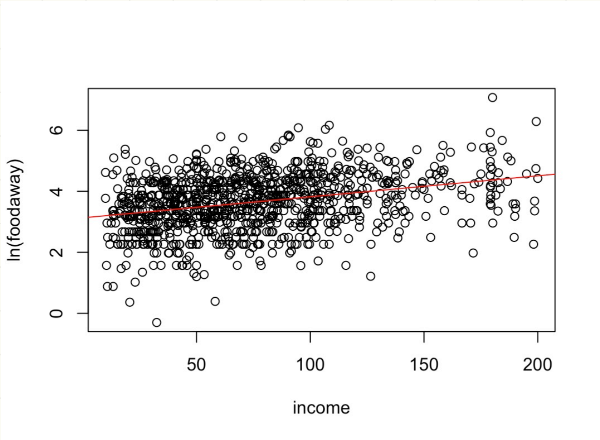
$$\widehat{\ln(foodaway)} = 3.1293 + 0.0069 \times income$$

$\beta_1 = 3.1293$. 當 income = 0 時. $\ln(foodaway) = 3.1293$

$\beta_2 = 0.0069$. 當 income 變動 1 單位時. ($100)

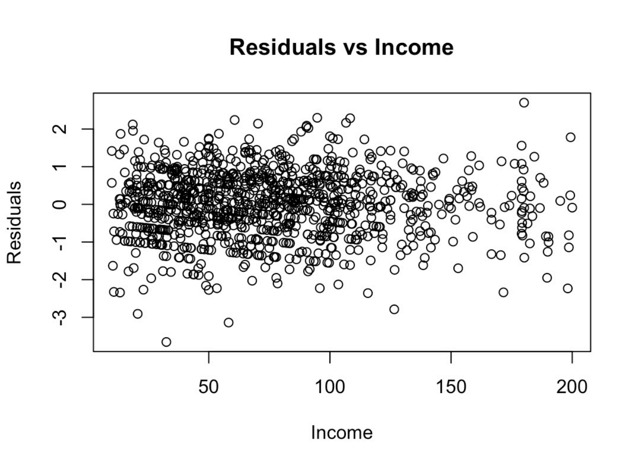
foodaway 變動 0.69%，且統計顯著.

(e)



可以看出ln(foodaway)和income呈正相關

(f)



OLS Residuals 看起來呈現隨機分佈，沒有特定的pattern.

但可以看出來所得越高觀測值越少.